

3.2.2 Group 2, the alkaline earth metals

The elements in Group 2 are called the alkaline earth metals. The trends in the solubilities of the hydroxides and the sulfates of these elements are linked to their use. Barium sulfate, magnesium hydroxide and magnesium sulfate have applications in medicines whilst calcium hydroxide is used in agriculture to change soil pH, which is essential for good crop production and maintaining the food supply.

Content	Opportunities for skills development
The trends in atomic radius, first ionisation energy and melting point of the elements Mg–Ba Students should be able to: - explain the trends in atomic radius and first ionisation energy - explain the melting point of the elements in terms of their structure and bonding. The reactions of the elements Mg–Ba with water. The use of magnesium in the extraction of titanium from TiCl_4 The relative solubilities of the hydroxides of the elements Mg–Ba in water. $\text{Mg}(\text{OH})_2$ is sparingly soluble. The use of $\text{Mg}(\text{OH})_2$ in medicine and of $\text{Ca}(\text{OH})_2$ in agriculture. The use of CaO or CaCO_3 to remove SO_2 from flue gases. The relative solubilities of the sulfates of the elements Mg–Ba in water. BaSO_4 is insoluble. The use of acidified BaCl_2 solution to test for sulfate ions. The use of BaSO_4 in medicine. Students should be able to: - explain why BaCl_2 solution is used to test for sulfate ions and why it is acidified.	AT c and k PS 2.2 Students could test the reactions of Mg–Ba with water and Mg with steam and record their results. AT d and k PS 2.2 Students could test the solubility of Group 2 hydroxides by mixing solutions of soluble Group 2 salts with sodium hydroxide and record their results. Students could test the solubility of Group 2 sulfates by mixing solutions of soluble Group 2 salts with sulfuric acid and record their results. Students could test for sulfate ions using acidified barium chloride and record their results. Research opportunity Students could investigate the use of BaSO_4 in medicine.